

Impact of the coronavirus disease 2019 pandemic on primary and metastatic lung cancer treatments in Japan: A nationwide study using an interrupted time series analysis

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ABSTRACT

Background: The coronavirus disease 2019 pandemic prompted healthcare providers to use different approaches from the current standards of care. We aimed to identify the changes in the number of treatments for primary non-small cell lung cancer (NSCLC) and metastatic lung cancer during the pandemic.

Methods: We used nationwide insurance claims data from January 2015 to January 2021, and estimated changes in the number of treatments using an interrupted time series analysis.

Results: The number of surgical resections for primary NSCLC significantly decreased in April 2020 (−888; 95% confidence interval [CI]: −1530 to −246) and July 2020 (−1314; 95% CI: −1935 to −694), while the number of stereotactic body radiotherapies (SBRTs) increased in April 2020 (95; 95% CI: 8–182) and July 2020 (111; 95% CI: 24–198). The total number of treatments for primary NSCLC remained unchanged; however, non-significant decreases were observed in 2020. The number of surgical resections for metastatic lung cancer significantly decreased in April 2020 (−201; 95% CI: −337 to −65), but it eventually increased in July 2020 (170; 95% CI: 32–308). Additionally, the number of SBRTs significantly increased in April 2020 (37; 95% CI: 3–71) and October 2020 (57; 95% CI: 23–91). The total number of treatments for metastatic lung cancer was maintained, with an initial decrease in April 2020 followed by a subsequent increase in July and October 2020.

Conclusion: In Japan, surgical triage for primary and metastatic lung cancer are likely to have been implemented during the pandemic. Despite these proactive measures, patients with primary NSCLC may have been untreated, likely owing to their undiagnosed disease, potentially leading to a deterioration in prognosis. By contrast, patients diagnosed with cancer prior to the pandemic are presumed to have received standard management throughout the course of the pandemic.

1. Introduction

The coronavirus disease 2019 (COVID-19) pandemic placed a tremendous burden on lung cancer healthcare providers, prompting them to use approaches that are different from the standards of cancer care. Cancer screening was suspended in the early phase of the pandemic

[1], and patients showing cancer symptoms may have been reluctant to seek care at outpatient clinics owing to fear of contracting the infection. Consequently, the number of cancer screenings, diagnostic examinations, and diagnoses for lung cancer decreased during the pandemic [2–10].

Previous studies conducted in various countries [4,11–15], including

Abbreviations: CI, confidence interval; COVID-19, coronavirus disease 2019; DPC, diagnosis procedure combination; MHLW, Ministry of Health, Labor and Welfare; NCD, the National Clinical Database; NDB, National Database of Health Insurance Claims and Specific Health Checkups; NSCLC, non-small cell lung cancer; PDT, photodynamic therapy; SARIMA, seasonal autoregressive integrated moving average; SBRT, stereotactic body radiotherapy.

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Japan [16–21], reported changes in the number of lung cancer surgeries during the pandemic. However, many studies conducted in Japan were single-institution studies [16], limited to specific regions [17,18], or questionnaire surveys with low response rates [19]. Therefore, these results may not accurately depict the impact of COVID-19 on lung cancer surgery in Japan as a whole. By contrast, the results of studies using the National Clinical Database (NCD) have high generalizability; the NCD contains data on more than 90% of all surgical procedures performed in Japan. Ikeda et al. reported that the number of lobectomy procedures in 2020 was lower than in 2018 and 2019 [20]. Sato et al. reported that the number of surgeries for primary lung cancer decreased in 2020, whereas that for metastatic lung cancer remained unchanged based on the trend over the past 5 years [21]. Several expert committees suggested postponing surgery in patients with non-fatal cancers and considering alternative treatment options to replace or delay surgery, such as stereotactic body radiotherapy (SBRT) and ablation, whenever feasible during the pandemic [22–25]. However, no study has reported the number of treatments other than surgery for lung cancer in Japan. Given the potential substitution of surgery for lung cancer with alternative treatments during the pandemic, changes in the number of possible alternatives, including SBRT and ablation, merit evaluation.

This study aimed to determine the impact of the COVID-19 pandemic on the treatment of primary and metastatic lung cancer including surgical resection, photodynamic therapy (PDT) or laser ablation, and SBRT in Japan. We used the National Database of Health Insurance Claims and Specific Health Checkups (NDB) of Japan, a nationwide database that contains comprehensive insurance claims data.

2. Methods

2.1. Data source

This was a repeated cross-sectional study using sample datasets from the NDB administered by the Ministry of Health, Labour, and Welfare (MHLW). Since April 2008, the NDB has stored all insurance claims data electronically submitted by various medical institutions to medical insurers. In January 2020, of the 50,336,611 medical insurance claims submitted, 49,463,250 (98.3%) were provided electronically and stored in the NDB. As of March 2020, 18,768 million insurance claims were accumulated in the database. The MHLW creates the sample datasets from the NDB in preparation for their distribution to third parties for research purposes. The features of the datasets are as follows: 1) datasets are created four times a year, in January, April, July, and October, and contain data for each month (one month's data); 2) data are randomly sampled from the NDB, with sampling rates of 1% for the medical outpatient claims and 10% for medical inpatient and diagnosis procedure combination (DPC) claims; and 3) personally identifiable information is excluded from the datasets. Sample datasets from January 2015 to January 2021 were obtained from the MHLW.

2.2. Target treatments and extraction procedures

Primary lung cancer only included non-small cell lung cancer (NSCLC) because the treatments for NSCLC and small cell lung cancer substantially differ. [Supplementary Material 1](#) shows the procedures for extracting the target insurance claims. The disease codes used for extraction are shown in [Supplementary Materials 2 and 3](#). For primary NSCLC and metastatic lung cancer treatments, insurance claims with disease codes corresponding to the International Classification of Diseases, Tenth Revision codes C34.X (malignant neoplasm of bronchus and lung) and C78.0 (secondary malignant neoplasm of the lung) were targeted, respectively. For primary NSCLC treatments, the codes for small cell lung cancer and rare cancer types, such as carcinoid, choriocarcinoma, and mucoepidermoid carcinoma, were excluded from the analysis. Targeted treatments for primary NSCLC and metastatic lung cancer were surgical resection, PDT or laser ablation, and SBRT. Surgical

resection was divided into four categories: sublobar resection via open surgery, lobectomy or pneumonectomy via open surgery, sublobar resection via thoracoscopic surgery, and lobectomy or pneumonectomy via thoracoscopic surgery. Sublobar resection included segmental and partial resections.

2.3. Creating time series data

Using the extracted insurance claims for each treatment, the time series data were created. First, we counted the number of claims for each insurance claim type (inpatient or DPC) for each month. The counted number was then multiplied by 10 for inpatient and DPC claims (multiplied by the reciprocal of the extraction rate). Finally, the total number of inpatient and DPC claims was calculated. Data on the outpatient claims were not used as the number of extracted claims was either zero or considerably small for each treatment. The time-series data are presented in [Supplementary Material 4](#).

2.4. Statistical analysis

An interrupted time series analysis was performed using the seasonal autoregressive integrated moving average (SARIMA) model based on the method introduced by Schaffer et al. [26], which was used in our previous studies [10,27,28]. All statistical analyses were performed using R version 4.2.0. First, we determined the SARIMA components, $(p, d, q) \times (P, D, Q)_s$, with the lowest Akaike's information criterion. Seasonality (s) was determined to be 4 as our time series data were created four times a year, in January, April, July, and October. In this step, all 25-point time series data were used, and the step-change variables during the pandemic were included in the model as external regressors. The step-change variables were assigned the values of 1 for each month during the pandemic and 0 otherwise, and are presented in [Supplementary Material 4](#). The period during the pandemic was defined as April 2020 to January 2021. In Japan, the first infected person was detected on January 15, 2020. Therefore, in January 2020, the impact on medical care was considered to be small or had not yet occurred. The estimates of the step-change variables represent the changes in numbers for each month during the pandemic. Second, the number during the pandemic was predicted using only the pre-pandemic points (21 points), which is a counterfactual number, assuming that the pandemic did not occur. In this step, the SARIMA components were applied to those determined in the first step. The rate of change was calculated by dividing the estimated change by the counterfactual number.

3. Results

We analyzed 20,452,831 medical outpatient claims, 3115,714 medical inpatient claims, and 2507,790 DPC claims. Since only 2 and 0 insurance claims were made for PDT or laser ablation for primary NSCLC and metastatic lung cancer, respectively, an interrupted time series analysis of the treatments was not performed. [Table 1](#) shows the changes in the number of treatments for primary NSCLC and metastatic lung cancer during the pandemic. The numbers of treatments for primary NSCLC and metastatic lung cancer are shown in [Figs. 1 and 2](#), respectively. The number of surgical resections for primary NSCLC significantly decreased in April 2020 (−888; 95% confidence interval [CI]: −1530 to −246) and July 2020 (−1314; 95% CI: −1935 to −694), while the number of SBRTs increased in April 2020 (95; 95% CI: 8–182) and July 2020 (111; 95% CI: 24–198) ([Fig. 1a and b](#)). The total number of treatments remained unchanged; however, non-significant decreases were observed in 2020 ([Fig. 1c](#)). [Supplementary Figure 1](#) shows the trends in the number of surgical resections performed by type of surgery. Thoracoscopic surgery was performed in most patients with primary NSCLC. The number of lobectomies or pneumonectomies via thoracoscopic surgery significantly decreased in July 2020, whereas the number of other types of surgical resections remained unchanged at that time.

Table 1

Estimated changes in the number and rate of treatments for primary non-small cell lung cancer and metastatic lung cancer during the COVID-19 pandemic.

		Estimated changes in number ²			Estimated changes in rate ³			
Treatments	Time	Number	95% CI		Rate (%)	95% CI		P-value
			Lower limit	Upper limit		Lower limit	Upper limit	
Primary NSCLC								
Surgical resection	2020/04	−888	−1530	−246	−20	−35	−6	0.007
	2020/07	−1314	−1935	−694	−29	−43	−15	<0.001
	2020/10	−343	−933	246	−8	−23	6	0.253
	2021/01	−635	−1476	207	−15	−35	5	0.140
SBRT	2020/04	95	8	182	31	3	60	0.032
	2020/07	111	24	198	36	8	64	0.012
	2020/10	4	−84	92	1	−28	31	0.932
	2021/01	−27	−121	67	−8	−35	19	0.574
Total treatments ¹	2020/04	−436	−1228	356	−10	−28	8	0.280
	2020/07	−439	−1265	387	−11	−30	9	0.298
	2020/10	−698	−1852	456	−15	−40	10	0.236
	2021/01	−32	−1232	1167	−1	−30	29	0.958
Metastatic lung cancer								
Surgical resection	2020/04	−201	−337	−65	−26	−43	−8	0.004
	2020/07	170	32	308	23	4	43	0.016
	2020/10	43	−90	177	5	−11	21	0.524
	2021/01	−121	−298	56	−15	−36	7	0.180
SBRT	2020/04	37	3	71	87	7	166	0.033
	2020/07	−3	−37	31	−7	−86	73	0.870
	2020/10	57	23	91	133	54	213	0.001
	2021/01	−23	−57	11	−53	−133	26	0.189
Total treatments ¹	2020/04	−112	−279	55	−14	−36	7	0.190
	2020/07	158	−14	331	21	−2	43	0.072
	2020/10	208	31	386	27	4	50	0.022
	2021/01	−52	−235	131	−7	−30	17	0.581

CI: confidence interval, NSCLC: non-small cell lung cancer, SBRT: stereotactic body radio therapy, COVID-19: coronavirus disease 2019

¹Surgical resection and SBRT were included.²Change in number for each month is presented.³Change in rate was calculated by dividing the estimated change in number by the counterfactual number.

The number of surgical resections for metastatic lung cancer initially decreased in April 2020 (−201; 95% CI: −337 to −65), but subsequently increased in July 2020 (170; 95% CI: 32–308) (Fig. 2a). Additionally, the number of SBRTs significantly increased in April 2020 (37; 95% CI: 3–71) and October 2020 (57; 95% CI: 23–91) (Fig. 2b). The total number of treatments remained unchanged in April 2020 but significantly increased in October 2020 (208; 95% CI: 31–386) (Fig. 2c). As shown in Supplementary Figure 2, the number of sublobar resections via thoracoscopic surgery significantly decreased in April 2020, but significantly increased in July 2020. Similarly, the number of lobectomies or pneumonectomies via thoracoscopic surgery non-significantly decreased in April 2020; however, it significantly increased in July 2020.

4. Discussion

Our findings revealed the changes in the number of treatments for primary NSCLC and metastatic lung cancer during the COVID-19 pandemic using nationwide insurance claims data in Japan.

The number of surgical resections for primary NSCLC significantly decreased in April and July 2020, with the largest decrease observed in July 2020. The reduction was mainly observed in thoracoscopic lobectomy or pneumonectomy. These results are consistent with previous studies using the NCD, which is a nationwide database that stores data of surgical procedures performed in Japan [20,21]. Ikeda et al. reported a significant decrease in the number of lobectomies in 2020 compared with the previous year [20]. Sato et al. reported that the number of surgeries for primary lung cancer decreased by 5.1% compared to 2019 and the greatest reduction was observed in July 2020 [21]. Additionally, the number of lobectomies for primary lung cancer decreased in 2020, but the number of segmentectomies and partial resections did not; meanwhile, the number of video-assisted thoracoscopic surgeries decreased, but that of open surgeries did not [21]. Our results enhance

the reliability of these findings, as consistent results were obtained using independent nationwide databases, such as the NDB and NCD. Additionally, we revealed changes in the number of treatments for primary lung cancer other than surgery, which cannot be determined by the NCD. Only 2 insurance claims with PDTs or laser ablations were extracted as treatments for primary NSCLC throughout the observation period, indicating that these treatments were rarely used in Japan. The number of SBRTs for primary NSCLC significantly increased in April and July 2020, when the number of surgical resections decreased, suggesting that surgical resection for early-stage NSCLC was partially replaced with SBRT. As a result, the total number of treatments for primary NSCLC remained unchanged during the pandemic owing to the decreased number of surgical resections and increased number of SBRT procedures performed. However, this number decreased in 2020, although the difference was not significant. Okuyama et al. reported that the number of lung cancer diagnoses detected by screening in 2020 decreased compared with the average number between 2016 and 2019 [2]. The decreased number of treatments we found may reflect a reduction in the number of diagnoses. Delayed treatment or surgical resection of NSCLC is associated with higher mortality [29,30]. Therefore, a poor prognosis in the future remains a concern.

Unlike primary NSCLC, the number of surgical resections for metastatic lung cancer decreased significantly in April 2020, followed by a significant increase in July 2020. This trend suggests that some surgeries scheduled in April 2020 were avoided and some were performed later (e. g., in July 2020). Additionally, the number of SBRTs significantly increased in April and October 2020, suggesting that SBRT was used to partially replace surgical resection for metastatic lung cancer. Meanwhile, the total number of treatments for metastatic lung cancer decreased in April 2020, but increased in July and October 2020. In summary, surgical resections for metastatic lung cancer were postponed or replaced with SBRT, and the total number of treatments was

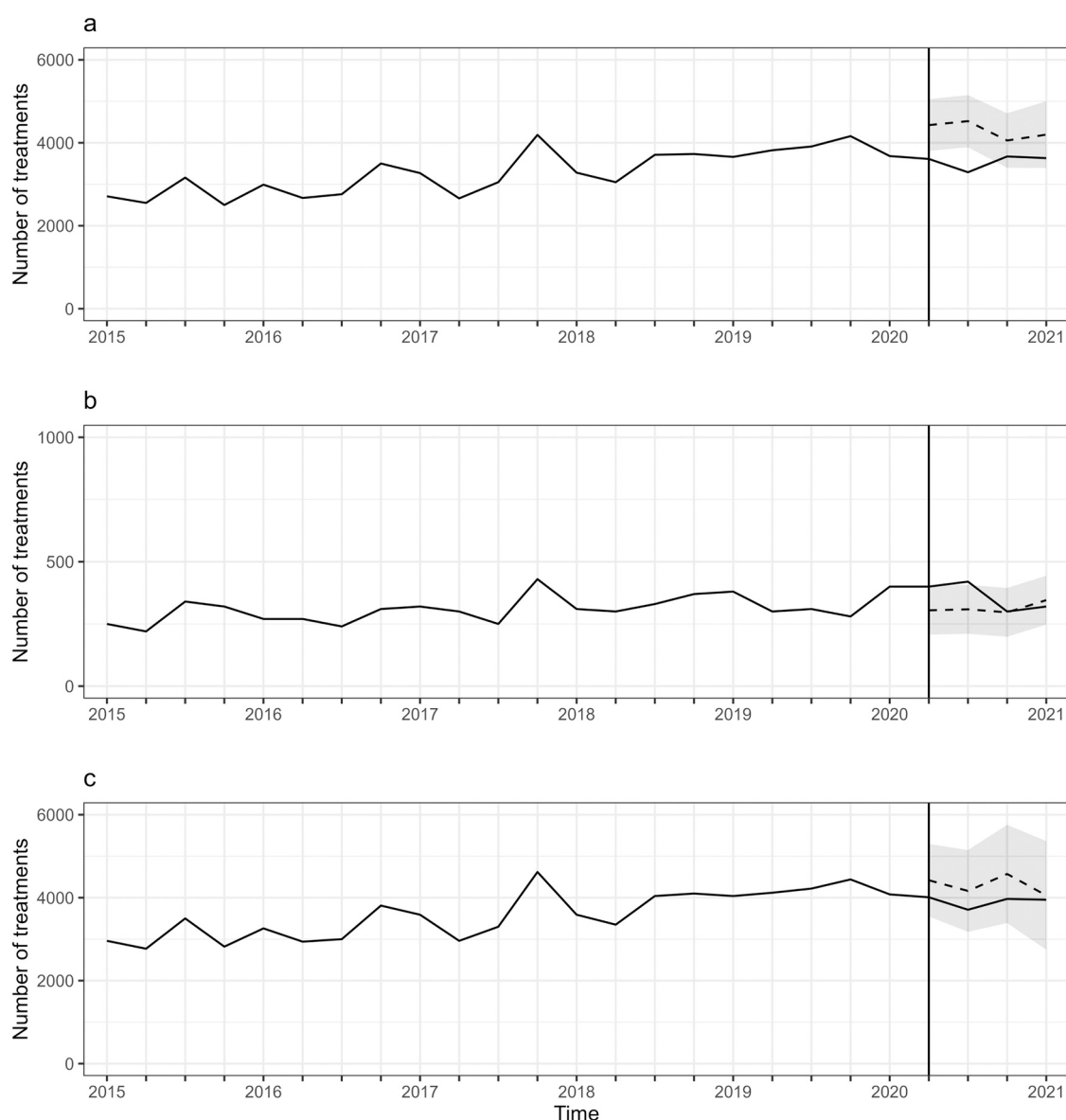


Fig. 1. Trends of the number of treatments for primary non-small cell lung cancer, a: Number of surgical resections, b: Number of stereotactic body radiotherapies, c: Total number of treatments, including surgical resection and stereotactic body radiotherapies. The observed numbers in all observation periods (solid line) and the expected counterfactual numbers with the corresponding 95% confidence intervals during the COVID-19 pandemic (dashed line and gray shading) are drawn. The data are presented four times a year: in January, April, July, and October.

maintained. Similarly, a previous study using the NCD reported that the number of surgical resections for metastatic lung cancer in 2020 followed a similar trend observed in the last 5 years [21].

Herein, we discuss the differences in trends between the total number of treatments for primary NSCLC and metastatic lung cancer. In Japan, the number of diagnoses for primary NSCLC decreased during the pandemic [2], owing to interrupted cancer screening [21] and a reduced number of diagnostic examinations performed in medical institutions [10]. Therefore, a reduction in the total number of treatments for primary NSCLC is thought to reflect a reduction in the number of diagnoses, which cannot be avoided by the efforts of health care providers in hospitals. By contrast, the diagnosis of metastatic lung cancer relies on the consistent performance of regular follow-up examinations for primary cancers. The total number of treatments, including surgical resections and other treatments, for metastatic lung cancer was maintained. Therefore, patients diagnosed with cancer before the pandemic would likely have continued to receive standard management during the pandemic.

The number of surgeries for lung cancer during the pandemic has been reported in various countries [4,11–15], including Japan [16–21]. Nonetheless, it remains uncertain whether surgical triage measures, such as treatment postponements or substitution with alternative therapies, were implemented during this pandemic. To our knowledge, only one study with a relatively small sample size reported an increased number of SBRTs performed during the pandemic, suggesting that SBRT was adopted to replace or postpone surgery [31]. Identifying the presence of untreated cancer patients is important for predicting future prognosis. However, if the number of untreated patients is determined by the number of surgeries, the impact of the pandemic may be overestimated as some of the surgical treatments may have been replaced with other treatments. This study is the first to demonstrate that surgical resection for NSCLC and metastatic lung cancer were replaced with SBRT during the pandemic in Japan using nationwide data. Our study also revealed that use of other treatments, especially possible alternatives to surgery, should be determined when assessing the impact of the pandemic on cancer care.

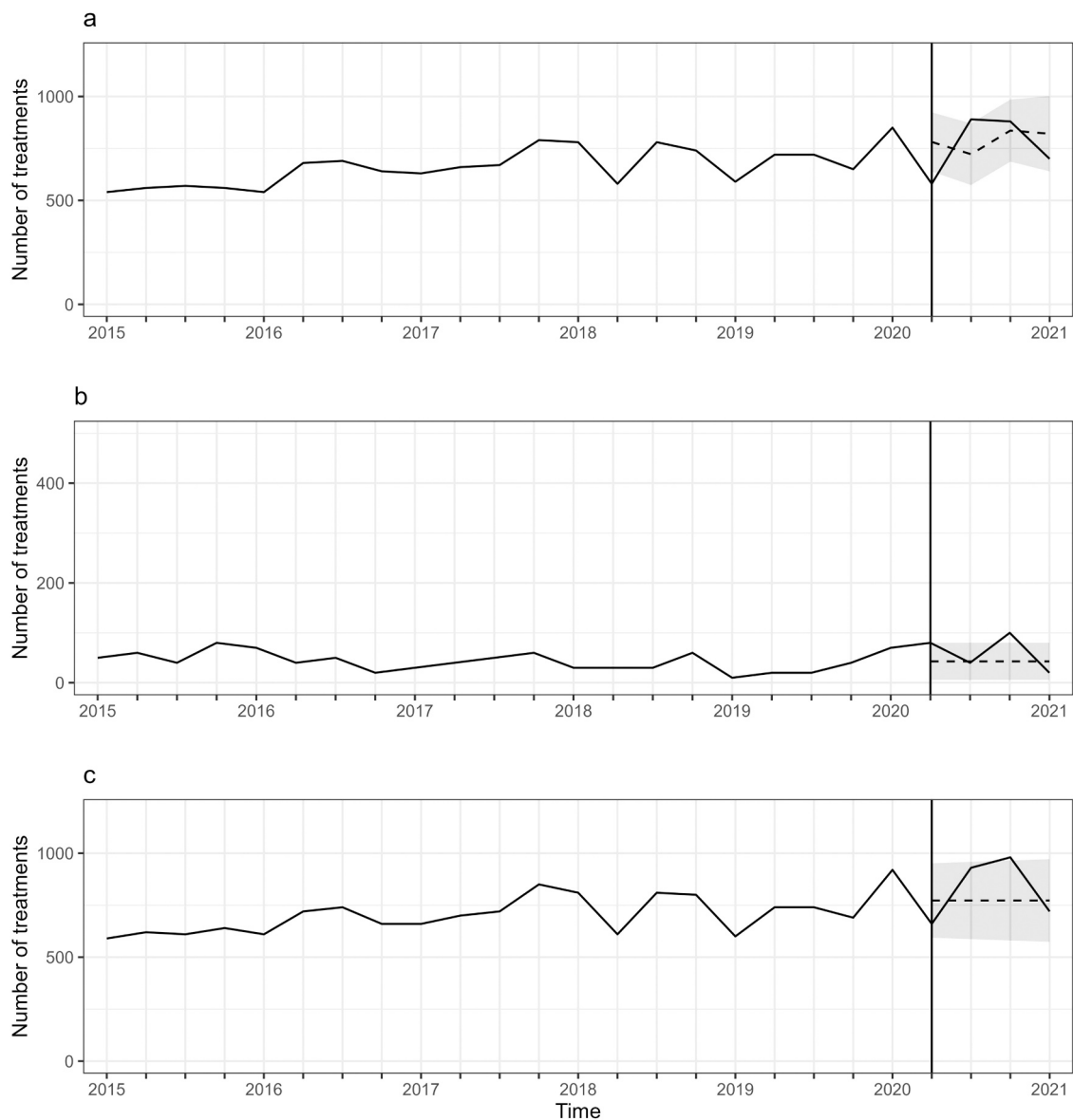


Fig. 2. Trends of the number of treatments for metastatic lung cancer, a: Number of surgical resections b: Number of stereotactic body radiotherapies, c: Total number of treatments, including surgical resection and stereotactic body radiotherapies. The observed numbers in all observation periods (solid line) and the expected counterfactual numbers with the corresponding 95% confidence intervals during the COVID-19 pandemic (dashed line and gray shading) are drawn. The data are presented four times a year: in January, April, July, and October.

SBRT is recognized as an attractive alternative to an invasive surgical procedure, because the risk of serious complications after SBRT is relatively low and three-year local control rates is approximately 90% for stage I NSCLC [32]. However, robust evidence comparing surgery and SBRT is lacking [32]. A pooled analysis of two randomized trials showed that overall survival at three years in SBRT was significantly higher than surgery [33]. By contrast, a meta-analysis comprising six studies conducted a propensity-matched comparison showed a superior three-year overall survival after surgery compared with SBRT [34]. Replacing surgery with SBRT during the pandemic could contribute to avoiding poor prognosis due to untreated disease. However, we cannot rule out the possibility that patients who underwent SBRT have a worse prognosis than those who underwent surgery.

This study has some limitations that warrant further consideration. First, we did not investigate the number of chemotherapies and external radiotherapy administered for lung cancer. We focused on curative treatments, considering the possibility of replacing surgery with alternative treatment options. However, chemotherapy and external

radiotherapy are used for curative and palliative care. We could not distinguish them because the sample datasets of the NDB are independent for each target month and do not include a purpose of use. Therefore, although surgery may have been replaced with chemotherapy and external radiotherapy, this study does not provide any indications in this regard. Furthermore, the reduction in the total number of treatments being overestimated is concerning because replacement of surgery with chemotherapy and external radiotherapy was not considered. Only patients with early-stage lung cancer are eligible for SBRT but a larger group of patients are eligible for surgery. Restricting the analysis to patients with early-stage cancer would improve accuracy in indicating the extent to which surgery replaced SBRT and the changes in the total number of treatments. However, this was not possible because insurance claims data did not include information on the stage of cancer. Second, the monthly trends could not be determined because the sample datasets were created four times per year. Third, the period of observation during the pandemic was too short to understand the potential long-term impact of the pandemic. However, we used the most recently

available sample datasets from the NDB. Fourth, random errors could occur, because we used sample datasets which were created by extracting randomly from the NDB. As a result, the confidence intervals became wider than those of the full dataset. Especially, for less frequent treatments, the inference of the error would be large.

Meanwhile, this study has several strengths. First, the trends in the number of treatments for primary as well as metastatic cancer provided additional implications for understanding cancer care during the pandemic. Second, the trajectories of the number of treatments other than surgical resection, such as SBRT, were investigated. To the best of our knowledge, this study is the first to suggest the implementation of surgical triage for lung cancer during the pandemic in Japan. Third, we used sample datasets from the NDB, which is a nationwide database containing comprehensive insurance claims data. Therefore, the study results have good generalizability. Fourth, we conducted an interrupted time series analysis using the SARIMA model. Most previous studies compared treatments in 2020 with those in 2019. Le Bihan-Benjamin et al. claimed that comparison with 2019 activities without consideration of the short- and medium-term trends leads to misunderstandings [35]. By contrast, we controlled for pre-existing trends, autocorrelation, moving average, and seasonality by an interrupted time series analysis using the SARIMA model, which is one of the best designs for establishing causality when randomized controlled trials are not feasible [26].

5. Conclusion

Surgical resections for primary NSCLC and metastatic lung cancer could have been partly postponed or replaced with SBRT. The total number of treatments for metastatic lung cancer remained unchanged; therefore, patients diagnosed with cancer before the pandemic are presumed to have received standard management throughout the course of the pandemic. The total number of treatments for primary NSCLC decreased in 2020, although the difference was not significant. This reduction would have been unavoidable with the efforts of healthcare providers in hospitals as it would reflect a reduction in the number of diagnosed cases. Undiagnosed and untreated primary NSCLC may lead to a poor prognosis in the future. As a valuable lesson from this pandemic, concerted efforts to avoid reducing the number of diagnoses are required.

Ethical statement

This study used sample datasets from the NDB, which were anonymized by the MHLW during their creation. Therefore, informed consent was not required. This study was approved by the Research Ethics Committee of the Chiba Foundation for Health Promotion and Disease Prevention (approval number: R3-4) and was conducted in accordance with the principles of the Declaration of Helsinki and Ethical Guidelines for Medical and Biological Research Involving Human Subjects.

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CRediT authorship contribution statement

Akira Hata: Writing – review & editing, Supervision, Methodology, Conceptualization. **Misuzu Fujita:** Writing – original draft, Visualization, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Hideyuki Hashimoto:** Writing – review & editing, Conceptualization. **Kazuya Yamaguchi:** Writing – review & editing, Conceptualization. **Yoshihiro Onouchi:** Writing – review & editing, Conceptualization. **Daisuke Sato:** Writing – review & editing, Conceptualization. **Takehiko Fujisawa:**

Writing – review & editing, Methodology, Conceptualization. **Kiminori Suzuki:** Writing – review & editing, Methodology, Conceptualization. **Kengo Nagashima:** Writing – review & editing, Methodology, Formal analysis, Conceptualization. **Tokuzo Kasai:** Writing – review & editing, Conceptualization.

Declaration of Competing Interest

The authors have no competing interests to declare.

Data availability

We used sample datasets from the National Database of Health Insurance Claims and Specific Health Checkups (NDB) provided by the Ministry of Health, Labour, and Welfare (MHLW) for this study. The MHLW prohibits the sharing of NDB data. Therefore, the minimal dataset underlying the results described in our manuscript cannot be shared. Those who want to access the sample datasets of the NDB must apply to the MHLW.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.canep.2024.102549](https://doi.org/10.1016/j.canep.2024.102549).

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